

Individuals at Risk for Vitamin A Toxicity

Individuals most at risk for vitamin A toxicity include patients taking systemic vitamin A derivatives for the treatment of dermatologic conditions such as **acne**, **psoriasis**, and **ichthyosis**. Other at-risk populations include vitamin food faddists who consume large quantities of nonprescription vitamin A supplements. Two notable episodes of vitamin A toxicity occurred in the 1950s, when very high levels of vitamin A were added to infant formulas, and in the 1970s, when high doses of vitamin A were used to treat various dermatologic diseases.

Clinical Findings

Acute Vitamin A Toxicity

- Dry, scaly skin
- Widespread desquamation
- Fissuring of the lips and angles of the mouth
- Headache, fatigue, anorexia, nausea, vomiting
- Blurred vision
- **Pseudotumor cerebri** (manifesting as headache and papilledema; infants may present with a bulging fontanelle)
- Myalgias and arthralgias

Chronic Vitamin A Toxicity

In Adults:

- Early sign: dryness of the lips
- Progression to: - Diffuse, dry, pruritic, scaly skin - Peeling of palms and soles - **Alopecia** - **Follicular hyperkeratosis** - Hyperpigmentation of the face and neck

- Systemic symptoms: anorexia, fatigue, weight loss

> **Note:** Follicular hyperkeratosis can occur in both **vitamin A deficiency (VAD)** and **vitamin A toxicity**.

In Children:

- Coarse hair with diffuse alopecia
- Coarse skin with generalized exfoliation
- Hyperpigmentation
- Exfoliative cheilitis
- Skeletal abnormalities: - Growth retardation due to premature closure of epiphyses - Spontaneous bone fractures

Proposed mechanisms for bone pathology include antagonism between vitamin A- and vitamin D-mediated intracellular signaling pathways and interactions with calcium-regulating hormones.

Long-term effects: - Deposition of excess vitamin A in adipose tissue - Perisinusoidal fibrosis of the liver, potentially progressing to **cirrhosis**

Laboratory Testing

Findings in hypervitaminosis A include: - Elevated serum **vitamin A** levels - Elevated **serum calcium** - Elevated **alkaline phosphatase (AP)**

These disturbances in calcium homeostasis may lead to: - Calcification of tendons - Ligament calcification - Soft tissue calcification

Liver involvement remains a major concern, with perisinusoidal fibrosis and potential progression to cirrhosis.

Treatment

- Discontinuation of excess vitamin A intake leads to resolution of **most symptoms**
- Exceptions:

- **Liver cirrhosis** (may be irreversible)
- **Consequences of pseudotumor cerebri** (e.g., permanent visual impairment)

Carotenemia and Carotenoderma

Epidemiology

- **Carotenemia** is a benign condition characterized by yellow-orange skin pigmentation due to elevated serum carotene levels.
- First described in 1919 by Hess and Meyers.
- More prevalent during World War I and World War II due to dietary shifts from meat-based to plant-based diets caused by food shortages.

Etiology and Pathogenesis

- Carotenes are not synthesized in the body and must be obtained from the diet.
- Dietary sources: carotene-rich fruits and vegetables (e.g., carrots, sweet potatoes, spinach).
- In the gastrointestinal tract:
 - Some carotene is converted to vitamin A (retinol)
 - Approximately one-third is directly absorbed unchanged

Factors affecting carotene absorption and metabolism: -

Hypothyroidism: Decreased conversion of carotene to retinol → elevated serum carotene - **Pancreatic lipase and bile acid deficiency** (due to pancreatic, biliary, or hepatic disease): Impaired digestion and absorption → increased carotene levels - **Food processing:** - Mashing, cooking, and pureeing increase carotene availability (rupture plant cell walls) - **Dietary fat:** Enhances absorption (carotene is fat-soluble) - **Dietary fiber:** Reduces absorption

Carotenoderma typically resolves with dietary modification and does not require specific treatment.